

VISIONINSPECT AI

Manufacturing Defect Detection & Quality Inspection System

Combined Milestone Report
Milestone 1 • Milestone 2 • Milestone 3

1. Project Overview

VisionInspect AI is an AI-powered manufacturing quality inspection platform designed to automatically inspect product images, detect defects and anomalies, classify quality issues, assess severity, and provide production-quality analytics. The reference project document defines the platform as a centralized system intended to reduce manual inspection effort, improve product quality, minimize manufacturing defects, and increase production efficiency.

The planned platform supports product image inspection, defect detection, quality-control automation, defect classification, manufacturing analytics, reporting, and related workflows.

2. Project Objectives

- Automatically detect product defects from images.
- Identify quality issues and anomalies using computer vision and AI.
- Support defect classification and severity assessment.
- Provide automated quality-control and pass/fail decisions.
- Generate inspection and production-quality reports.
- Maintain inspection statistics and manufacturing analytics.
- Provide a centralized platform for inspection workflows.

3. Reference System Modules

Module	Purpose
User Management	Quality engineer/factory supervisor accounts, authentication, authorization and role management.
Image Acquisition	Product image upload, camera-integration simulation, batch processing and image validation.
Image Processing	Preprocessing, noise removal, enhancement and feature extraction.
Defect Detection	Defect identification, anomaly detection, object detection and defect localization.
Defect Classification	Defect type classification, severity scoring, categorization and confidence analysis.
Quality Control	Pass/fail decisions, inspection reporting, recommendations and production monitoring.
Manufacturing Analytics	Defect trends, production-quality reports, inspection statistics and operational insights.

4. Technology Stack Referenced

The reference document identifies Python/FastAPI for the backend; React.js/Next.js for the frontend; PostgreSQL/MongoDB for databases; OpenCV, TensorFlow, PyTorch, YOLO, CNN models, Scikit-learn, NumPy and Pandas for AI, computer vision and data processing; Docker/AWS/Azure for deployment; Git/GitHub for version control; VS Code as the IDE; and Postman for API testing.

5. Milestone 1

5.1 Milestone Objective

Milestone 1 focuses on project initialization, design, core setup, workflow definition, authentication, image inspection management and dataset integration.

5.2 Reference Requirements

- Define quality-inspection workflows and project objectives.
- Design system architecture and database schema.
- Create UI wireframes and workflow planning.
- Set up frontend and backend environments.
- Implement authentication and role-based access.
- Load the MVTec AD dataset.
- Build image-upload workflows and an inspection dashboard.

5.3 Work Performed

The project foundation was established around the manufacturing inspection workflow. The MVTec AD dataset was selected and loaded as the industrial anomaly-detection dataset. Image acquisition, dataset loading and preprocessing formed an important part of the data preparation work. The project was organized into frontend, backend, model and inspection-related components.

My primary contribution in this milestone was image acquisition, dataset loading and preprocessing. This included preparing the image data so that it could be consumed by the later model and inspection pipeline.

5.4 Milestone 1 Workflow

Project initialization → Dataset acquisition → Image loading → Image organization → Preprocessing → Model-ready image/data → Inspection workflow

5.5 Milestone 1 Outcome

- Project structure and core development environment established.
- MVTec AD dataset integrated into the project workflow.
- Image acquisition/loading and preprocessing workflow established.
- Foundation created for subsequent anomaly detection and inspection modules.

5.6 Mentor Explanation — Milestone 1

“In Milestone 1, our main focus was establishing the foundation of VisionInspect AI. We defined the manufacturing inspection workflow, selected and loaded the MVTec AD dataset, and prepared the image data through the preprocessing stage. My main responsibility was image acquisition, dataset loading and preprocessing. The output of this milestone was a structured, model-ready data workflow that could be used in the next milestone.”

6. Milestone 2

6.1 Milestone Objective

Milestone 2 focuses on image processing, defect detection, model training/integration, defect prediction, image analytics and inspection monitoring.

6.2 Reference Requirements

- Implement image preprocessing pipelines.
- Generate image quality-analysis reports.
- Build image analytics workflows.
- Train defect-detection models.
- Generate defect predictions.
- Build inspection monitoring dashboards.

6.3 Work Performed

The project moved from data preparation to AI-based anomaly/defect detection. An autoencoder-based anomaly-detection approach was used in the project workflow. The inference pipeline processes an input image and produces an anomaly-related result, which is then used to determine whether the product is good or defective.

The backend was implemented using FastAPI so that inspection functionality could be exposed through APIs. The project included prediction, inspection and related backend modules, allowing the model output to be consumed by later severity, statistics and reporting components.

6.4 Defect Detection Workflow

Input image → Preprocessing → Anomaly detection/model inference → Anomaly-related score → Decision/threshold → Good or Defective

6.5 Autoencoder Concept

An autoencoder learns to encode an input and reconstruct it. In an anomaly-detection workflow, the model can learn the characteristics of normal images; reconstruction differences can then provide an anomaly signal. The project uses this concept as part of the defect-detection pipeline.

6.6 Backend/API Integration

FastAPI was used to expose the inspection functionality. Swagger UI provides an interface for testing the backend endpoints. The project workflow included inspection, prediction and related modules, supporting the transition from model inference to a complete inspection application.

6.7 Milestone 2 Outcome

- Image-processing and anomaly-detection workflow established.
- Defect prediction pipeline implemented.
- FastAPI backend used to expose inspection functionality.
- Prediction outputs prepared for integration with severity, statistics and reporting.

6.8 Mentor Explanation — Milestone 2

“In Milestone 2, we moved from data preparation to actual defect detection. The image is preprocessed and passed through the anomaly-detection pipeline. The system produces an anomaly-related result and determines whether the product is good or defective. We also exposed the inspection functionality through FastAPI, which prepared the backend for integration with the later severity, statistics and reporting modules.”

7. Milestone 3

7.1 Milestone Objective

Milestone 3 focuses on defect categorization, severity scoring, quality assessment, production-quality reporting, defect analytics and trend monitoring.

7.2 Reference Requirements

- Implement defect categorization workflows.
- Generate severity-scoring reports and quality-risk assessments.
- Build quality-assessment modules.
- Generate production-quality reports.
- Build defect-analytics dashboards.
- Develop trend-monitoring workflows.

7.3 Work Performed

The project was extended from basic defect prediction into a broader inspection-intelligence workflow. The recent implementation work included severity assessment, inspection statistics, report generation and integration of these components with prediction results.

The severity module provides a score and maps it to a severity category. The statistics module tracks inspection counts and derives good/defective totals and pass/failure rates. The reporting component collects inspection information into a report. These modules were connected to the inspection workflow.

7.4 Severity Scoring — Reference Framework

The reference document defines a weighted severity framework based on four parameters:

Parameter	Weight	Meaning
Defect Size	30%	Physical size of the defect relative to the product surface.
Defect Location	25%	Whether the defect is in a critical or non-critical area.
Defect Type	25%	Seriousness of the detected defect category.
Detection Confidence	20%	Model confidence in the defect prediction.

Reference formula: $\text{Severity Score} = (\text{Size} \times 30\%) + (\text{Location} \times 25\%) + (\text{Defect Type} \times 25\%) + (\text{Confidence} \times 20\%)$.

7.5 Reference Severity Levels

Level	Score	Reference interpretation
Critical	80–100	Major structural defect; product

		rejection/immediate action.
High	60–79	Significant quality issue; repair or rework recommended.
Medium	40–59	Moderate quality concern; inspection review required.
Low	0–39	Minor cosmetic defect; product generally acceptable.

Important implementation note: the reference framework is the planned severity specification. The exact formula and thresholds should be claimed as implemented in the mentor presentation only if they are present in the current project code.

7.6 Inspection Statistics

- Total images inspected
- Good products
- Defective products
- Pass rate
- Failure rate

The statistics module converts individual inspection results into aggregate quality information. This supports monitoring of inspection outcomes and provides a basis for manufacturing analytics.

7.7 Inspection Reporting

The reporting workflow consolidates inspection information such as prediction, severity-related information and inspection statistics into a quality-inspection report. The purpose is to provide a structured result that can be reviewed rather than requiring users to interpret raw API responses.

7.8 Milestone 3 End-to-End Workflow

Upload image → Preprocessing → Defect/anomaly prediction → Severity assessment → Inspection statistics → Quality/inspection report

7.9 Milestone 3 Outcome

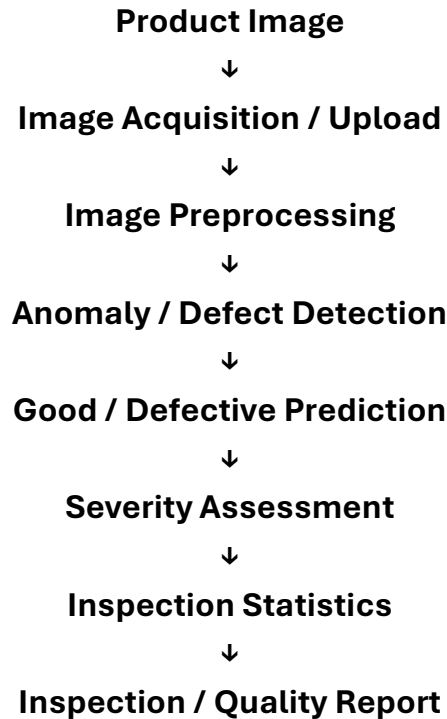
- Prediction integrated with severity assessment.
- Inspection statistics module integrated.
- Inspection/report generation workflow developed.
- Backend inspection modules connected into an end-to-end workflow.
- Foundation established for quality analytics and future trend monitoring.

7.10 Mentor Explanation — Milestone 3

“In Milestone 3, we extended the basic defect-prediction system into a more complete quality-inspection workflow. Instead of only returning Good or Defective, the system can provide severity-related information, maintain inspection statistics, and generate inspection reports. These modules were integrated with the prediction pipeline so that the output of one stage can be used by the next stage.”

8. Combined End-to-End Architecture

The combined workflow developed across the first three milestones can be explained as:



9. Current Demonstration Plan

For a mentor demonstration, the project should be prepared so that the FastAPI server is already running before the live demonstration. The presenter can open the Swagger documentation directly and demonstrate the inspection workflow without spending presentation time on environment setup.

1. Start the backend in VS Code before the meeting and verify that it runs successfully.
2. Open the FastAPI Swagger documentation page.
3. Use one previously tested demo image.
4. Execute the inspection endpoint and explain the returned prediction.
5. Execute the severity functionality and explain the score/category.
6. Execute the statistics functionality and explain the aggregated values.
7. Show report generation if it is working in the current implementation.
8. Keep screenshots of successful outputs as a backup.

10. Suggested Mentor Presentation Sequence

Section	What to explain
Introduction	Project title, problem and objective.
Milestone 1	Dataset, project setup, image acquisition/loading and preprocessing.
Milestone 2	Anomaly detection, prediction pipeline and FastAPI backend.
Milestone 3	Severity, statistics, reporting and integration.

Architecture	Explain the complete image-to-report workflow.
Live Demo	Open Swagger directly and execute a prepared inspection.
Results	Show actual outputs from the current implementation.
Challenges	Explain integration, preprocessing and model/pipeline issues.
Future Work	Accuracy validation, better severity calibration, analytics, deployment and testing.

11. Suggested Live-Demo Script

“Now I will demonstrate the working inspection workflow. The backend has already been started, so I will directly open the Swagger documentation. I will upload a prepared product image and execute the inspection endpoint. The image goes through preprocessing and the anomaly-detection pipeline, after which the system returns the inspection result. I will then show how the severity information is generated, how the inspection statistics are updated, and how the result can be used for reporting.”

12. Key Questions to Prepare For

Mentor Question	Suggested Answer
Why MVTec AD?	It is an industrial anomaly-detection dataset suitable for normal/anomalous product-image inspection.
Why an autoencoder?	It can learn normal-image representations and use reconstruction differences as an anomaly signal.
What is your contribution?	My primary contribution is image acquisition, dataset loading and preprocessing, with additional work on inspection-module integration.
Why FastAPI?	It exposes the inspection functionality as APIs and supports communication between backend AI services and application components.
What does the severity score mean?	It is an algorithmic quality/severity indicator; it should not be described as a direct physical measurement unless validated against physical defect measurements.
What is the current limitation?	The system needs broader validation, threshold/calibration work and evaluation against real manufacturing quality data before production use.

13. Overall Milestone Summary

Milestone	Focus	Outcome
Milestone 1	Foundation	Project setup, architecture/workflow planning, MVTec AD integration, image acquisition/loading and preprocessing.
Milestone 2	Detection	Image processing, anomaly

		detection, model inference, defect prediction and FastAPI inspection workflow.
Milestone 3	Inspection Intelligence	Severity assessment, statistics, reporting and integration into an end-to-end quality inspection workflow.

14. Conclusion

Across Milestones 1–3, VisionInspect AI progressed from project and data preparation to AI-based defect detection and then to a broader inspection-intelligence workflow. The project foundation was established in Milestone 1, the anomaly-detection and prediction pipeline was developed in Milestone 2, and severity, statistics, reporting and integration were addressed in Milestone 3. The resulting workflow provides a foundation for the later testing, validation, deployment and documentation activities defined in the reference plan.

15. Source Basis

This combined report is based primarily on the uploaded reference document, “VisionInspect AI: Manufacturing Defect Detection & Quality Inspection System.” The milestone structure, requirements, severity framework, technology stack, evaluation criteria and performance goals are preserved from that reference. Implementation descriptions are phrased conservatively where the reference describes planned requirements rather than verified implementation details.